

Automated Item Generation with Generative and Predictive ML

Fall School, University of Tübingen — Day 1: Framework, history, generation

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Session 1A — Framing and the pre-LLM history of AIG

background

psychometrics

machine learning

education

other

show of hands

Python without help?

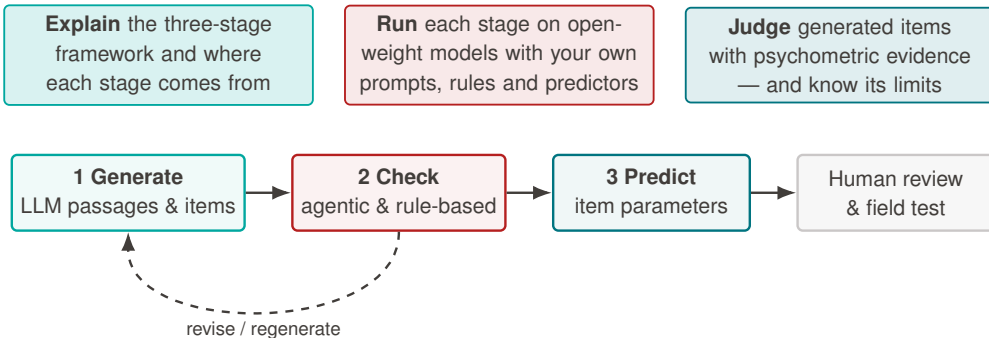
fitted an IRT model?

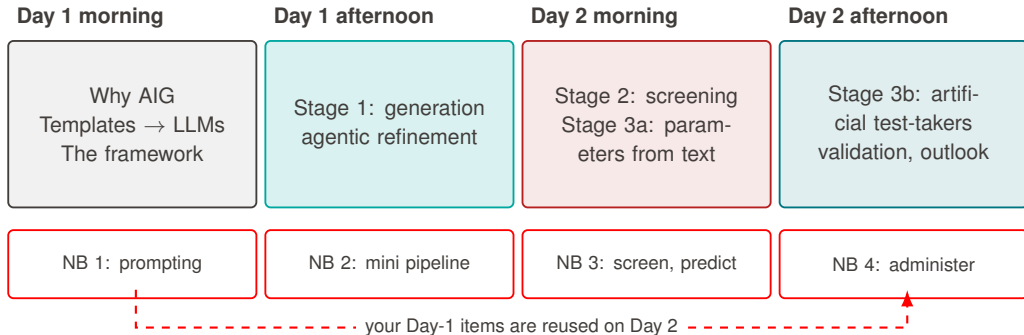
wrote a real test item?

called the LLM API from code?

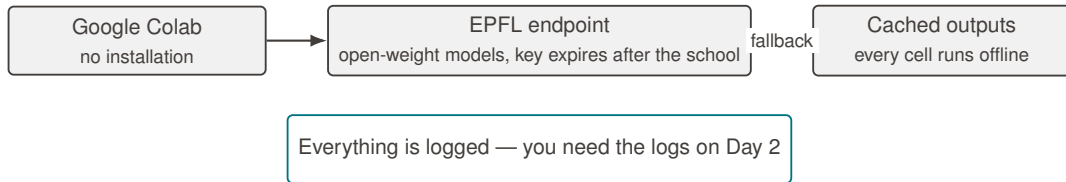
Why we ask

Two days of mixed psychometric and ML material. Knowing where you are lets us skip what you know.

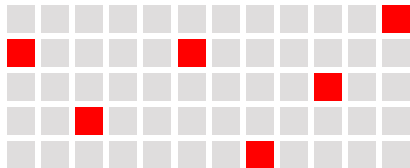




Coffee 10:30–10:45 and 14:15–14:30 · Lunch 12:00–13:00



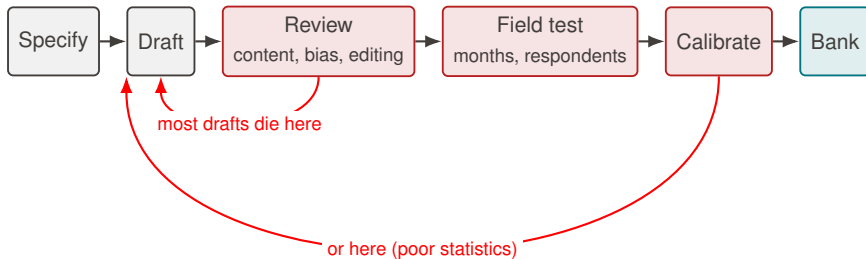
Work in pairs; mixed backgrounds are an advantage.



calibrated item bank (grey) vs. one adaptive test (red)

- PISA, PIRLS, licensure, admissions, language certification
- Items are exposed, leaked, retired — and replaced
- Adaptive and continuous testing multiply demand

The calibrated bank is the scarce asset, not the test form.



Every arrow costs expert time. Note *where* the cost sits: review and calibration, not drafting.

Reliability

discriminates between test-takers

Validity

measures the construct,
not reading load or trivia

Fairness

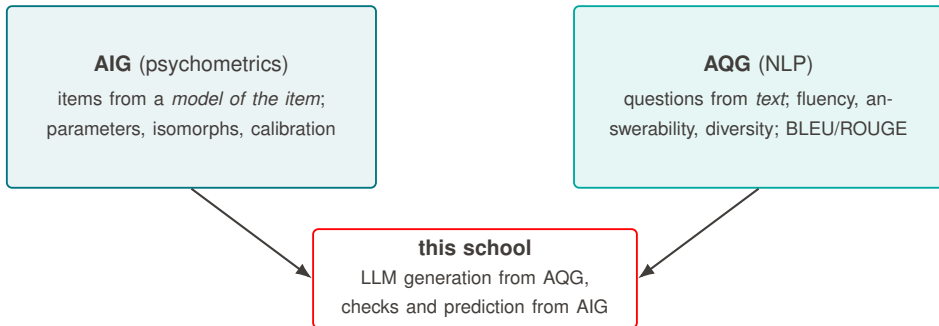
no DIF, no
loaded content

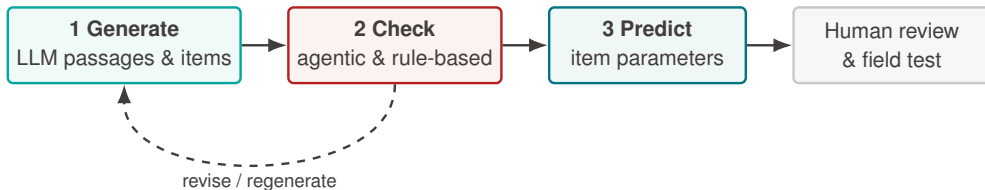
Craft

no item-writing flaws

cheapest to check →
first screen (Day 2)

Any generator must satisfy all four. The framework checks them from cheap to expensive.





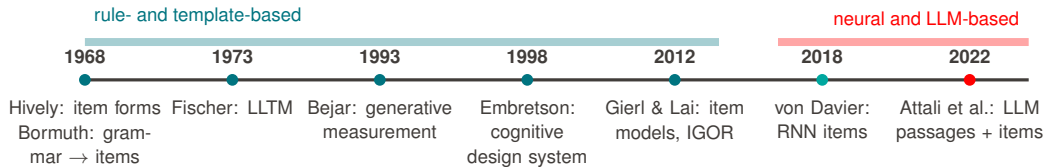
Generate: passages and items, guided by assessment frameworks and few-shot exemplars

Check: critique–rewrite–rate loops; rule-based screens for flaws, sensibility, solvability

Predict: 2PL parameters from text; keep only items in the target range

Developed by R. Jensen at ETH Zurich: semester project (Jensen 2025, supervised by R. Debelak) and master's thesis (Jensen 2026, advised by M. Sachan and R. Debelak).

Sixty years of AIG in twenty minutes



A shop reduces the price of a [item] from [P] to [Q]. What is the percentage decrease?

replacement set {jacket, bicycle, phone, ...}

numeric slots $P \in \{40, \dots, 120\}$, Q constrained by P

Hively, Patterson & Page (1968): a fixed shell with replacement sets; every item is a sample from a defined universe.

Bormuth (1969): questions from prose via transformational grammar — the first text-to-item method.

Validate the specification, not the instance.

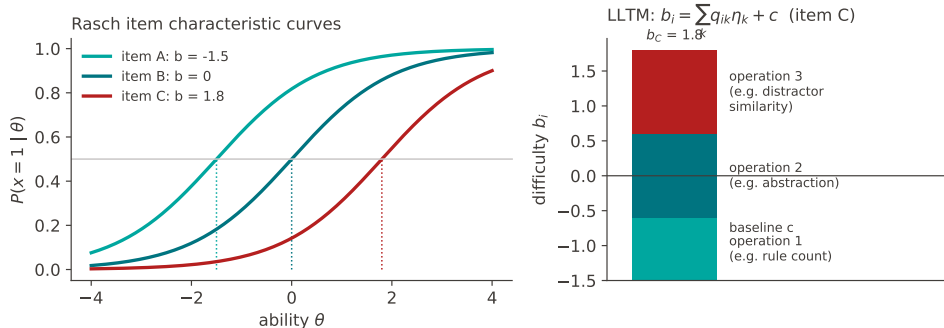
trivial or un-
natural items

no difficulty control

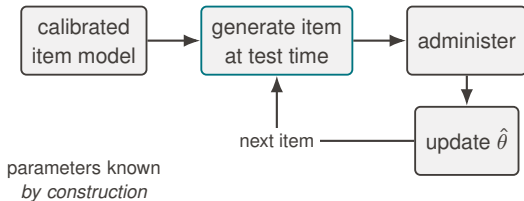
surface features
drive statistics

no infrastructure to
deliver or calibrate

all four return with LLM-generated items — keep the list

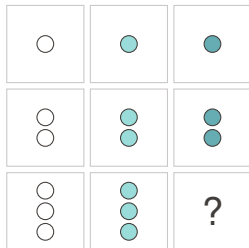


Fischer's LLTM (1973): Rasch difficulty decomposed into weighted cognitive operations. If it holds, a *new* item's difficulty follows from its features — the warrant for strong-theory AIG and the ancestor of Day 2's text-based predictors.



Bejar (1993): items are instances of a model whose parameters are established — the field test moves from the item to the item model.

Bejar et al. (2002/2003): feasibility of on-the-fly generation inside adaptive testing (GRE quantitative).



schematic: two rules (count, shading); difficulty rises with number and type of rules

- Start from a cognitive theory of the task
- Design features manipulate processing demands (rule count, rule type, perceptual load)
- Predict difficulty from features (LLTM), then confirm empirically

Embretson (1998): design features explain a large share of difficulty variance for matrix items.

Arendasy & Sommer: Rasch-calibrated generators for figural matrices and other reasoning tasks, used operationally in high-stakes assessment. Batches calibrated and re-checked for fit and DIF.

radicals

should change
parameters
reasoning steps,
distractor type

incidentals

should not
names, word-
ing, option order

test: items differing only in incidentals
share parameters (*statistical isomorphism*)

Irvine & Kyllonen (2002)



1-layer models vary elements in a fixed stem; n -layer models also vary structure and context · weak theory reproduces a good item; strong theory builds from a cognitive account of difficulty

Gierl & Lai, ITEMS module, Educational Measurement (2013)

Stem

A shop reduces the price of an item from [P] to [Q]. What is the percentage decrease?

Elements

$P \in \{40, 50, 60, 80, 120\}$; $Q = P(1 - r)$,
 $r \in \{0.10, 0.20, 0.25, 0.40\}$

Key: $100r$

Distractors, misconception-coded

- $100(P - Q)/Q$ wrong base
- $P - Q$ absolute difference
- $100Q/P$ remaining fraction

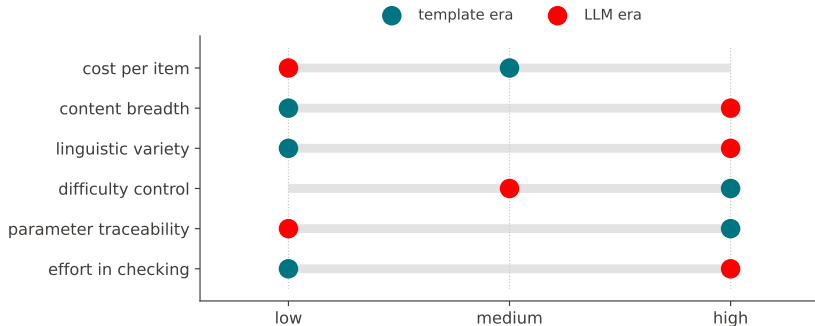
Constraints: integer key; distractors distinct, positive.

0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
---	---	---	---	---	---	---	---	---	---	----	----	----	----	----	----	----	----	----	----

radicals: r , base · incidentals: shop, item, currency

$5 \times 4 = 20$ isomorphs

What templates buy and what they cost



Templates solved stage 3 for constrained domains and never touched stage 1 for open text (passages).

parent-item calibration

siblings inherit parameters; verify on a sample

statistical isomorphism

siblings share a , b within error

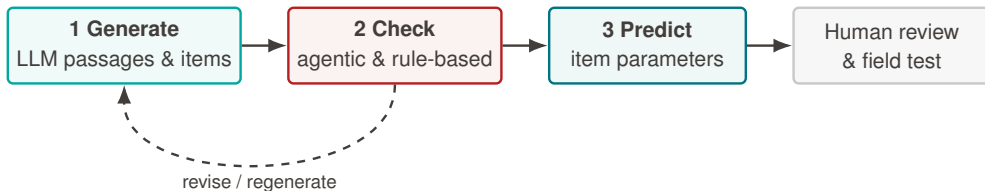
expert review

content validity, craft

Carry forward

Day 2 asks the same three questions of LLM-generated items — with fewer guarantees, because there is no explicit item model to inherit from.

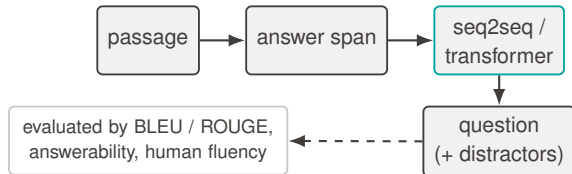
Which framework stage did template-based AIG already solve, and which did it not touch?



Coffee — back at 10:45

Session 1B — The neural era and the framework

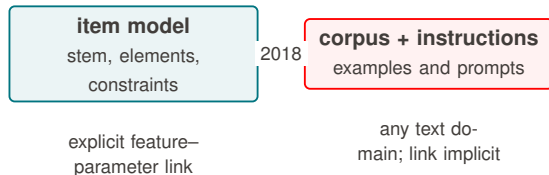
From item models to language models



Kurdi et al. (2020): hundreds of systems; Du et al. (2017) and successors for neural QG; distractor generation for RACE-style exams.

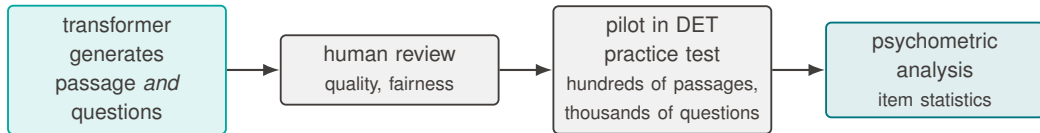
Missing

Item parameters. Fluent, answerable, diverse — and still a bad test item.



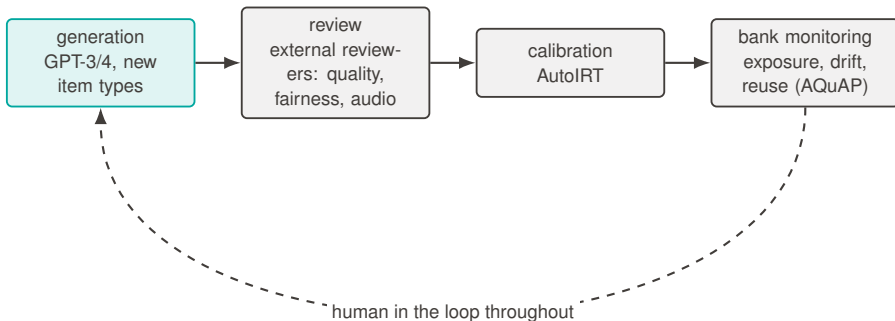
von Davier (2018): an RNN trained on personality items produces new statements.

“Training Optimus Prime, M.D.” (2019): GPT-2 on medical items — plausible, not guaranteed correct.



Why it matters

First demonstration at scale that an LLM-based pipeline produces items that survive both expert review and empirical item analysis. The validation pattern — review *and* data — is the one we adopt.



Generation is one box of four. Most of the engineering is in review, calibration and monitoring.

instruction only
“write a text for
12-year-olds”

weaker level match

instruction + 2–3
exemplar passages
at the target level

better level match

Bezirhan & von Davier (2023): few-shot passage generation with target grade or age.

Exemplars are *inspiration* for style and level, not templates for substitution; the output is a new text.

Jensen (2025) §4.2

Passage (excerpt): "...the bridge, completed in 1890, was the longest of its kind in Europe ..."

Q. The bridge, completed in 1890, was the longest of its kind in which continent?

A. Asia B. Africa **C. Europe, where it remained the longest of its kind for decades** D. Antarctica

1 stem copies the passage verbatim

2 answerable without the passage

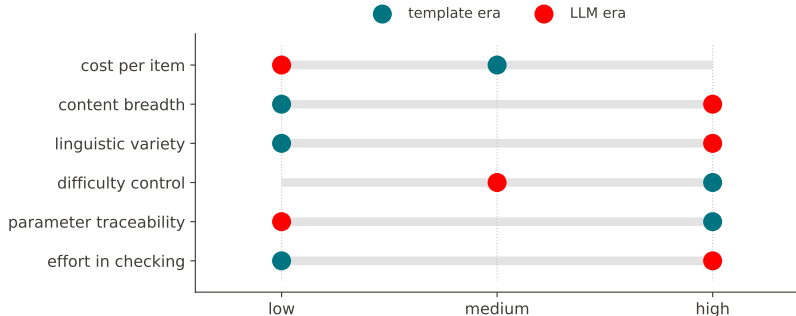
3 key is longest and most specific

4 distractors implausible

Plus: factual errors in informational passages (5) and difficulty clustering at the easy end (6; Jensen 2026).

Compare with the 1968 list.

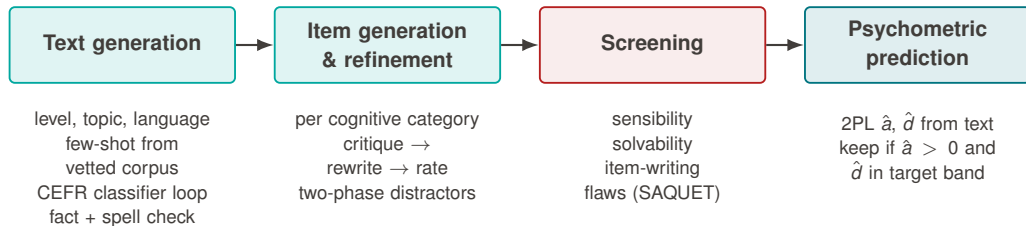
Säuberli & Clematide (2024); Sayin & Gierl (2024)



The hinge of this school

Generation became cheap. Quality assurance became the bottleneck. That is why the framework has stages 2 and 3.

A holistic framework for AIG



Jensen (2025); Jensen (2026) §3.1

grounded

categories follow
PIRLS/PISA,
not the model's
taxonomy

separated

the writer
does not judge

checkable

every stage emits
rationales, ratings,
flaw vectors,
parameters

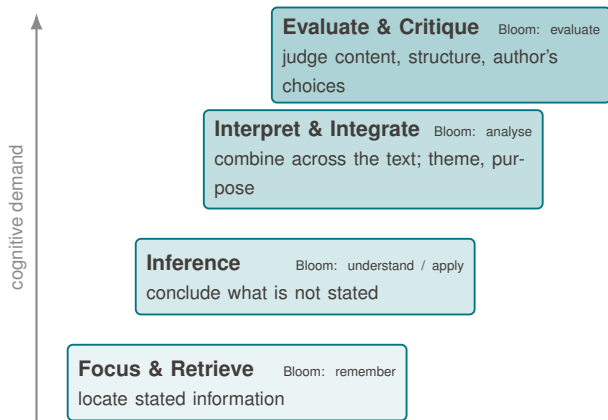
specialised

narrative vs.
reasoning models

**human at
the end**

automation moves
the review point

distilled from Jensen (2025, 2026)



Derived from PIRLS/PISA reading processes; each category has its own prompt, rubric and constraints. Jensen (2026) §3.2

FR: stem and key
paraphrase; no
shared keywords

INF: key follows from
the text but is not in it

INT: key needs at least
two passage locations

CRIT: key defensible
from the text, not taste

These constraints do the work that elements and constraints did in Gierl & Lai's item models. The template is now a prompt — and the checks must verify that the constraints were honoured.

Jensen (2026) §3.2

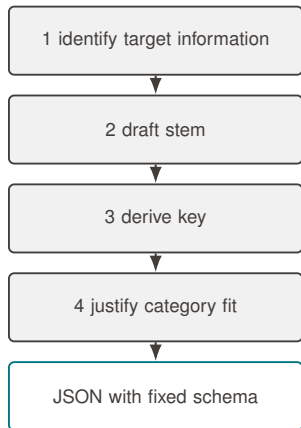
Reasoning model — today: Qwen3.x thinking

- + structured, step-by-step deduction
- + item generation, misconception-based distractors, rubric rating
- verbose; can lose to standard models on simple tasks

Flagship model — today: Apertus

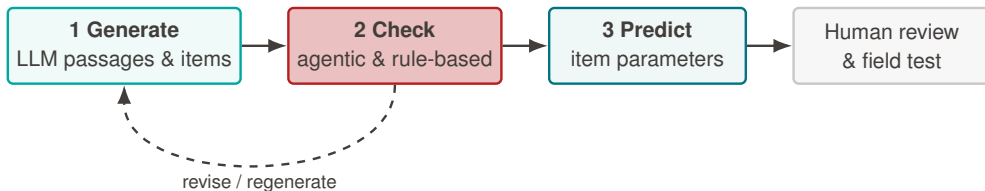
- + broad knowledge, narrative fluency
- + passages, dialogues, stories
- + token-efficient on low-complexity tasks
- weaker at constraint-following

Jensen (2025) §4.3



```
{"category": "INF",  
  "stem": "...", "key": "...",  
  "rationale": "...",  
  "passage_spans": [[12,48],[130,171]]}
```

The rationale is stored with the item: evidence for review, and a debugging aid when a check fails.
Jensen (2025) §4.4



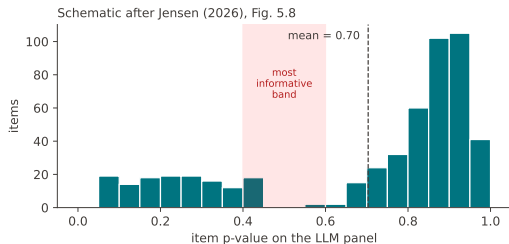
Agentic (this afternoon)
critique → rewrite → rate, until
threshold or cap

Distractors (this afternoon)
overgenerate, then filter

Rule-based (tomorrow)
sensitivity, solvability, SAQUET's
flaw rubric

Jensen (2025) §5–6; Jensen (2026) §3.1

- encoders + linguistic features $\rightarrow$ RF / boosting $\rightarrow \hat{a}, \hat{d}$
- keep if $\hat{a} > 0$ and $\hat{d}$ in the target band
- human item bank: $r \approx .78$ predicted vs. empirical



Tomorrow's twist

On LLM respondents the same predictor gives $r = -.08$, and items pile up at the easy end (plot: schematic after Jensen 2026, Fig. 5.8).

Jensen (2025) §7.2; Jensen (2026) §3.3, §5.4, ch. 8

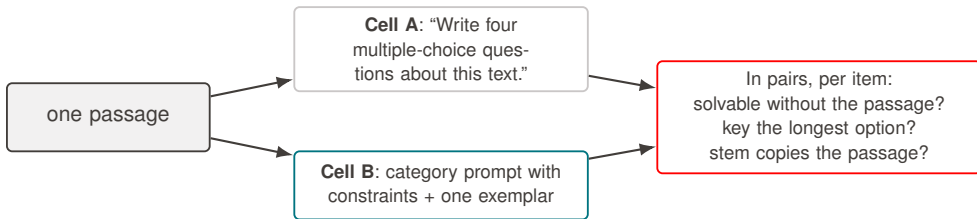
no replacement
for field testing
prediction is a filter

no fairness review
sensitivity, DIF stay
human and data-driven

text only
no images, au-
dio, interaction

no guarantee on
literary or opin-
ion passages

Expert reviewers in the semester project (Jensen 2025, §8) still found items to edit after all automated stages. The stages reduce the reviewers' load; they do not remove the reviewers.



Keep your notes — we compare with the automated checks tomorrow.

Notebook by R. Jensen

- What did the naive prompt get wrong most often?
- Did the constrained prompt fix it, or trade it for a different problem?
- Which of the six failure modes did you *not* see?

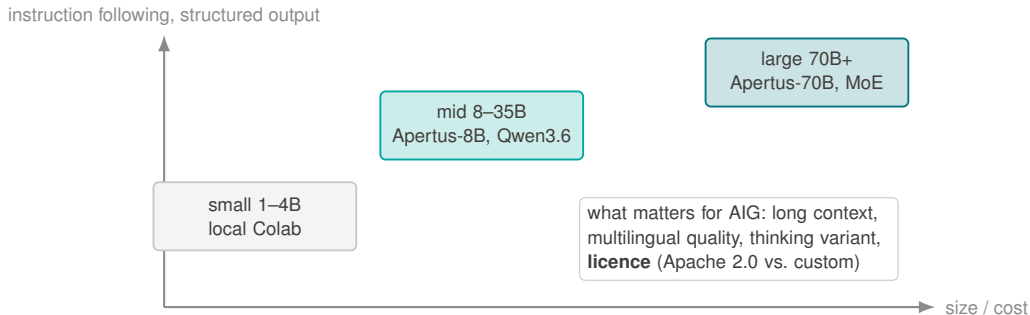
This afternoon we build the loop that catches these automatically.

Lunch — back at 13:00

Session 2A — Stage 1: Generation with open-weight models

Building stage 1

The open-weight landscape for this task



Reasoning traces can run to thousands of tokens per call — budget for it.

Apertus — EPFL / ETH Zurich / CSCS,
2025

8B and 70B · Apache 2.0

fully open: weights, *training data*, recipe

role today: passage generation

documented training data = the only basis for
reasoning about item contamination

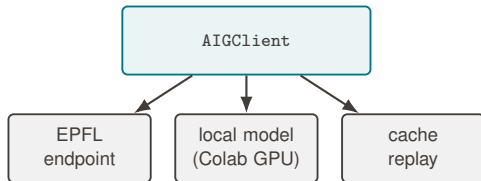
Qwen3.x — Alibaba, Apache 2.0 tier

mid-size instruct / thinking variant

role today: item generation, distractors, judging

thinking on for generation and rating, off for
cheap checks

both served from EPFL via one OpenAI-compatible endpoint · lo-
cal fallback: small Qwen3.x or quantised Apertus-8B on Colab



```
client = AIGClient(  
    base_url=ENDPOINT, api_key=KEY,  
    text_model="apertus-70b",  
    reason_model="qwen3-thinking",  
    cache_dir="cache/")  
  
out = client.chat(  
    model="reason", messages=msgs,  
    json_schema=ItemSchema,  
    temperature=0.7, seed=1)
```

Switching to a local model changes `base_url` only; every call is cached and logged.

temperature

0.7–1.0 for
passages and
distractor pools;
0–0.2 for judging

seeds

fix for repro-
ducibility; vary for
isomorph families

schema

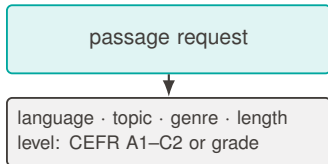
JSON / con-
strained decoding;
validate be-
fore storing

logging

model, prompt
hash, params,
tokens, output
— per call

stop conditions

cap iterations
and retries

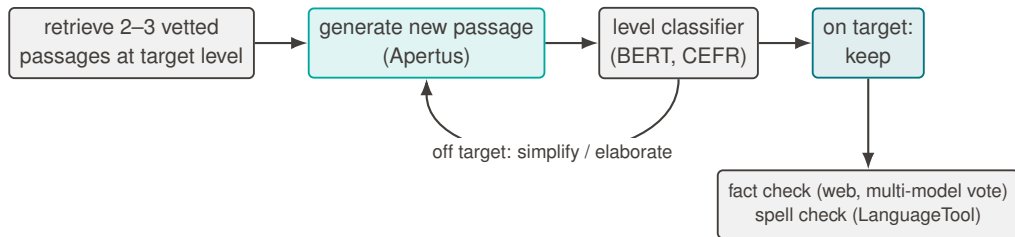


informational: a good item must be unanswerable from general knowledge

literary: solvable from the passage *with* interpretation

Genre decides the solvability check downstream.

Jensen (2025) §7.1; Jensen (2026) §3.1



The classifier is the “radical” control: it enforces level independently of the generator’s self-report. Fact-checking only for informational text.

Jensen (2026) §3.1, stage 1

B1 passage
same topic

readability indices, classifier verdict

C1 passage
same topic

one loop iteration if off target

Ask the room: which sentence would you cut to make the C1 text B2?

1 role and construct

2 category definition + constraints

3 rubric the item is rated against

4 passage

5 one exemplar of
the same category

6 reasoning steps + output schema

Output: stem, key, rationale, passage spans.

Why the rubric goes in

The generator optimises what it is shown;
showing it the rater's rubric aligns the two.

Jensen (2025) §5; Jensen (2026) §3.1

helps

- + explicit negative constraints (“do not quote the passage”)
- + one exemplar per category, not five
- + asking for spans forces grounding
- + schema-first output

breaks

- long stems that restate the passage
- keys that are the only complete sentence
- “all / none of the above”
- stem, key and distractors in one shot

Distractors are generated separately, after the stem and key are stable. lessons from building the pipeline: Jensen (2025)

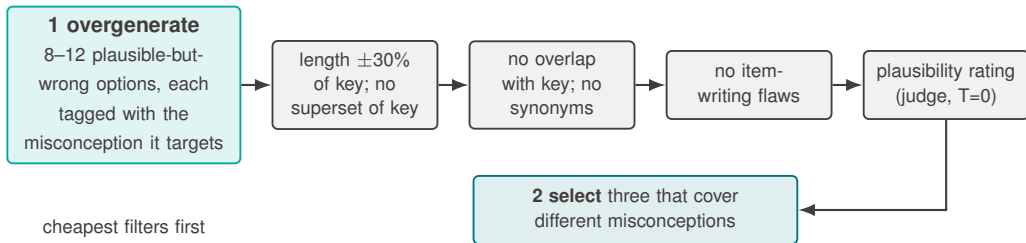
template supplies

numbers in ranges and constraints
solution path and key
one misconception per distractor

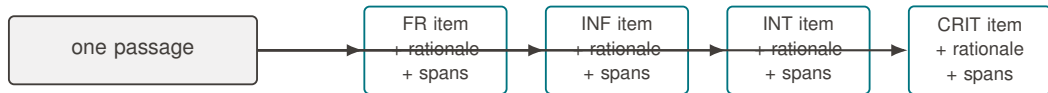
LLM supplies

context (shop, discount, unit), natural phrasing
distractor wording consistent with the context
no numbers — fixed by the template

difficulty stays controllable (radicals in the template); variety comes from the model (incidentals). Eedi's diagnostic items are the real-world model of misconception-coded distractors.



Open question (Sachan group, 2026): can LLMs model *incorrect* student reasoning well enough that real students pick the distractors? Tomorrow's empirical check. Jensen (2025) §6.3; Feng et al. 2024; Scarlatos et al. 2024

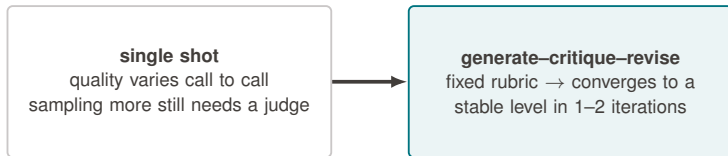


Spot one flaw together — there is usually one. Note how INF and CRIT differ in what the rationale must justify.

Coffee — back at 14:30

Session 2B — Agentic refinement and Notebook 2

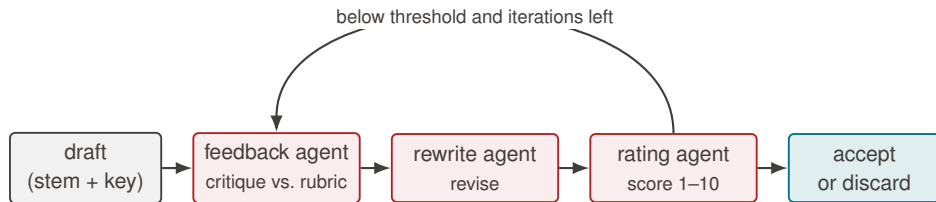
Generation is cheap; judging is the work



LLM-as-judge applied *upstream*, during construction, rather than at scoring time.

Jensen

(2026) ch. 1



Stop at threshold or cap (cap = 2 in Notebook 2). Distractors only for accepted stem–key pairs.

Jensen (2025) §5; Jensen (2026) §3.1

INF rubric (example)

key not stated in the passage but follows from it
2

exactly one defensible inference 2

stem does not reveal the key; no verbatim text 2

language at or below passage level 2

answerable in under a minute 2

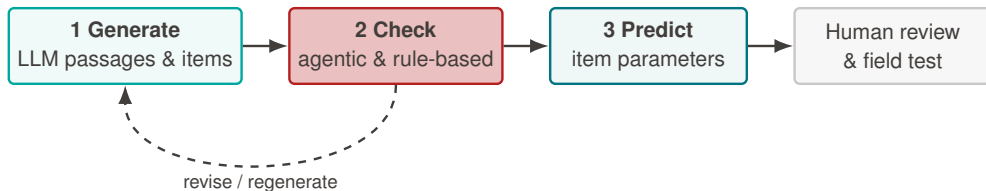
Same rubric shown to the generator and used by the rater (rubric design after Jensen 2025, §5; the lines shown are an example).

Separation

Rater and generator should be different models. Same model, different prompt is a weak separation.

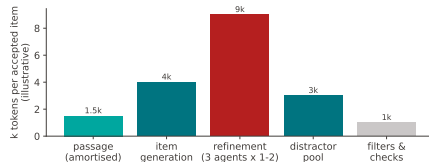
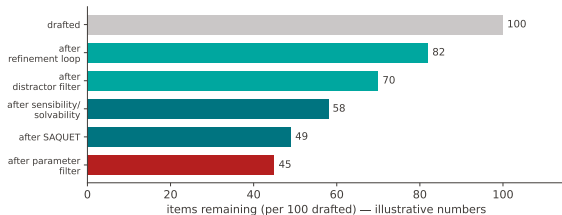
Known bias	Mitigation
position bias	swap orders; rate single items, not pairs
verbosity bias	length-controlled rubric lines
self-preference	different model as rater
rubric drift across items	fixed rubric, temperature 0, multiple raters, voting
overconfidence	calibrate against a small human-rated set

Ratings are measurements and need their own validity check. Literature: Zheng et al. (2023) and follow-ups.



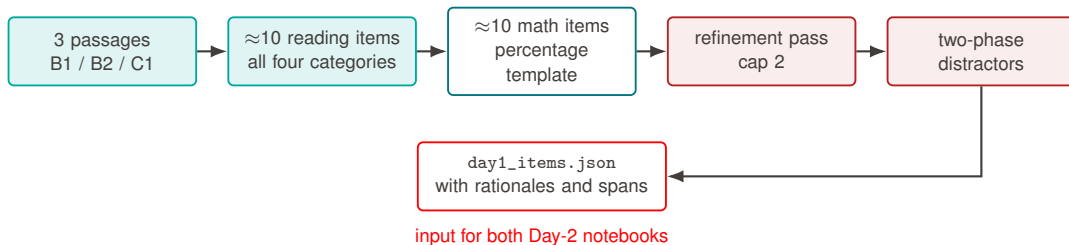
the loop optimises the item as text: category fit, clarity, grounding

it does not check: sensibility and solvability against the passage, flaws across the option set, predicted parameters — Day 2



Cut in this order when the budget is tight:
iteration cap, thinking mode on cheap checks,
distractor pool size.

The psychometric filter removes few items once earlier stages work (Jensen (2025) §8).



Notebook sections: client → passages → items → loop → distractors → export. Prompts live in one cell each — edit there.

Notebook by R. Jensen

raise the pass score
 $7 \rightarrow 8$; count iterations and rejections

swap the reasoning model for the small local one; compare rejection rates

change one rubric line;
watch the rewrite agent

three isomorphs from the math template with different contexts
— keep for tomorrow



Two pairs, one item each: what changed, and was the change an improvement?

item forms $\rightarrow$ item models $\rightarrow$ language models

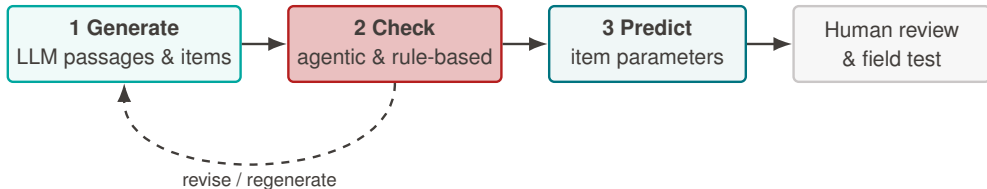
LLTM: features explain difficulty

the three-stage framework and its principles

stage 1 end to end, with an agentic loop

Things you saw go wrong

- ... (filled in live)



Morning: item-writing flaws and SAQUET; sensibility and solvability; parameters from item text (CMCQRD). Notebook 3 on your `day1_items.json`.

Afternoon: LLMs as artificial test-takers; what transfers between human and model respondents; validating the whole framework. Notebook 4.

Optional reading: Moore et al. (2024), SAQUET; Liu, Bhandari & Pardos (2025), LLM respondents.

pace

too slow / right / too fast

level

too basic / right
/ too advanced

one thing to change
for tomorrow

See you tomorrow at 9:15